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Evaluation of cybersecurity awareness among high school students in Turkey before and after training

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Abstract

The rapid expansion of digital technologies has increased young users' exposure to online threats and has made the need for effective cybersecurity education at the secondary school level more apparent. This study examines cybersecurity awareness among high school students and evaluates a structured training program implemented in four different types of high schools in the Darica district of Turkey. Data were collected with a five-point Likert-type survey before and after the training. The 601 students who were present in both measurements formed the dataset used for the pre-post analysis. The awareness scale was examined through principal component analysis which yielded a five-dimensional structure with acceptable overall internal consistency (Cronbach's alpha = 0.86). Individual-level change was analyzed with paired-samples t-tests and Cohen's d. Differences by school, grade and gender were tested with analysis of variance and independent-samples t-tests. Finally, students' ability to distinguish genuine from fraudulent website screenshots was examined with McNemar tests. Before the training, male students and students in academically selective and technically oriented schools showed higher awareness on several dimensions. After the training scores for action-oriented protective behaviors rose, while self-reported scores on the conceptual-knowledge dimensions declined. One possible interpretation is that students formed a more realistic view of their own knowledge rather than losing it, though this was not tested directly. Recognition of several fraudulent pages improved, for example from 50.4% to 63.9% on one item, although accuracy on the genuine pages did not increase. On some dimensions differences between schools and between genders were smaller after the training, while on others they persisted. Since the study compared matched participants without a control group these results are interpreted as changes associated with the training rather than as definitive proof of cause. Overall, the findings indicate that standardized, school-wide cybersecurity training was associated with changes in students' protective behavior and in their recognition of several fraudulent pages and they support integrating such programs into secondary education.

Clinical trial number: Not applicable.

Keywords Cybersecurity awareness, Secondary education, Cybersecurity education/Training, Digital safety, Information literacy

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1 Introduction

The proliferation of internet utilization has facilitated profound advancements in communication technologies. However, it has simultaneously introduced a wide array of security challenges. User-induced vulnerabilities frequently serve as the initial point of compromise, as malicious actors systematically exploit psychological triggers, including trust, fear, authority, and urgency to manipulate individuals [1, 4]. Commonly employed tactics include phishing, pretexting, baiting and scareware, all designed to deceive targets into divulging passwords or other sensitive information often by leveraging cognitive biases and emotional responses [27]. The motivations underpinning these social engineering schemes are multifaceted and may span financial gain, unauthorized access to confidential resources, ideological objectives, personal vendettas or even the impulse to disrupt systems or introduce disorder.

Moreover, social media platforms have emerged as especially fertile vectors for social engineering efforts, offering attackers rich repositories of personal and behavioral data. By monitoring online activities, adversaries compile detailed profiles on potential victims which enable them to craft highly tailored phishing communications or deceptive phone interactions [3]. Among digitally savvy younger populations, who maintain elevated levels of social media engagement, enhancing critical digital literacy and raising user awareness have proven to be pivotal in mitigating risk and enhancing resilience against social engineering threats.

According to the most recent report issued by the Virtual Private Network (VPN) provider SurfShark, 110.8 million accounts were compromised globally during the second quarter of 2023. In terms of the number of breached accounts, Turkey ranked fifth (Surfshark, [28]). Furthermore, at the Cyber Security Weekend META 2024 conference convened by the cybersecurity firm Kaspersky in Kuala Lumpur, Turkey was identified as the nation with the highest proportion of users (41.8%) impacted by online threats. In summary, within the META region-comprising the Middle East, Turkey, and Africa-Turkey was identified as the country with the highest share of users affected by online threats (Kaspersky, [13]).

Most studies of cybersecurity awareness treat awareness as a single, self-evident quality that a person either possesses or lacks. This, however, makes it difficult to clarify what changes a training intervention should bring about in an individual. To address this gap, the present study draws on two behavioral models that are widely used in information security research: Protection Motivation Theory and the Health Belief Model. These two models help explain why a person adopts or ignores recommended protective behaviors. In doing so, they treat awareness not as a single piece but as a set of separate beliefs that can change through education.

Protection Motivation Theory, proposed by Rogers [25], is a socio-cognitive model that explains how individuals respond to threats and to warning messages. According to the theory, a person carries out two appraisals before acting. The first is threat appraisal, which concerns how serious a threat is and how likely it is to affect the person. The second is coping appraisal, which concerns whether the recommended behavior actually works (response efficacy) and whether the person believes they are able to perform it (self-efficacy). Johnston and Warkentin, [11] applied this reasoning to computer users and found that these appraisals shaped users' intention to comply with security recommendations. Seen this way, cybersecurity training is effective when it raises students'

perception of risk while convincing them that simple measures, such as using strong passwords or recognizing phishing, are both useful and within their reach.

The Health Belief Model, in turn, is a psychological theory that explains why individuals do or do not engage in behaviors that protect their health or prevent illness, and it offers an account that complements the previous model. Originally developed in public health to explain preventive behavior, the model proposes that people act when they believe they are susceptible to a problem, that its consequences are serious, that a particular behavior is beneficial, and that the barriers in front of that behavior can be overcome; the behavior is often initiated by an external trigger known as a cue to action [26]. Ng et al., [21] adapted the model to computer security and showed that perceived susceptibility, perceived benefits, and self-efficacy predicted whether users handled email attachments safely. Within this framework, a structured training session can be regarded as a deliberate cue to action that also reinforces the underlying beliefs.

Building on these two models, the study defines what the administered survey was used to measure and what the training was expected to change. The instrument was designed to examine some of the models' dimensions, such as students' awareness of threats and their seriousness, their knowledge of protective measures and their perception of the benefits of using them, and their confidence in applying these measures in practice. The standardized training is therefore treated as an intervention aimed at shifting these specific beliefs rather than at raising a single undifferentiated awareness level. This perspective also helps interpret a pattern that appears in the results. Despite the practical importance of school-based cybersecurity education, few studies evaluate such a program against a theoretically grounded, multidimensional definition of awareness, particularly in the context of Turkish secondary schools. The present study addresses this gap.

2 Literature review

The increasing number of cyberattacks indicates that traditional education and awareness methodologies are still inadequate for building the necessary cybersecurity skills and capabilities. While employees at some organizations receive specific security training, security awareness for the vast majority of technology users such as ordinary people, including students, is limited to security tips found on certain websites. As a result, most security awareness programs fail. The rising number of data breaches and other cyberattacks clearly demonstrate that these tips are not enough to raise public security awareness to the level required to create a secure cyber culture [23]. Recent research confirms this inadequacy, highlighting the issues, such as the requirement for specialized programs, easy access to resources, and uniform evaluation standards [37]. Continuous efforts to improve cybersecurity education and awareness are crucial as the digital ecosystem changes. These programs are essential to protecting people, businesses, and countries from the constantly changing threats of the digital era in a world where information really is power [19]. Furthermore, innovative pedagogical strategies such as gamification and virtual labs have shown measurable improvements in learning outcomes, engagement, and self-regulated cybersecurity skill development [14, 32].

Many academic studies have been conducted on the cybersecurity awareness of high school students. For example, in Turkey, Yılmaz et al., [35] examined the internet security and computer usage awareness profiles of middle school students in Bartin. They

found that most students had inadequate information security and computer usage awareness and concluded that the students may be at risk from online threats. They provided feedback to parents, schools, and policymakers to increase information security and computer usage awareness among students [37].

Tosun et al., [31] held a workshop in Istanbul to discuss cybersecurity and the proper use of social media. The workshop discussed the current issues related to cybersecurity and social media use in Istanbul and a SWOT analysis for resolving these issues, offering some potential solutions. SWOT analysis is a widely used management method to identify and analyze an organization's resources and environmental factors in four dimensions: Strengths, Weaknesses, Opportunities, and Threats [31]. The workshop's report highlighted the students' lack of perception regarding cybersecurity and underscored how social media can be used to raise awareness about it. Therefore, it is important to draw attention to these issues.

Looking at international studies, Rahman et al., [24] state the need for cybersecurity education by drawing attention to the potential risks faced by young individuals, including cyberbullying, online fraud, and other harmful activities. Research on technology-enhanced education also offers guidance for designing and evaluating such interventions. For example, Kuvvetli and Kazu , [17] examined how AI-supported instruction affected learning outcomes, illustrating approaches to rigorously assessing technology-integrated education. More recent studies show that the rise of AI-driven cyberattacks has made the integration of cybersecurity education into digital literacy curricula even more urgent (World Economic Forum [WEF], [34]). Recent work also emphasizes integrating generative AI into cybersecurity courses to enhance critical thinking, policy awareness, and responsible AI use [9]. These threats require students to possess not just technical skills, but also critical thinking abilities to protect themselves from the misuse of AI. In this context, education becomes a proactive measure against such threats, providing technical skills and fostering a sense of empowerment among students.

In their research, [22] propose a new framework to help combat human error, one of the most popular weaknesses in cybersecurity, by providing information security awareness and training programs through gamification, known as Cyber-Hero. This framework helps transfer a set of security instructions to be learned in a serious game that students in schools can engage with at an early stage. This approach is supported by recent studies. Research by Tempestini et al., [29] suggests that game-based learning, when combined with video and text-based training, significantly increases cybersecurity awareness among young adults. Similarly, Williams et al., [33] demonstrated that game-based strategies can successfully engage even non-technical students, increasing their interest and retention of cybersecurity concepts [18] further found that structured gamification programs significantly improve students' security awareness and motivation. This hands-on, problem-solving method is considered more effective for retaining complex information than traditional approaches.

Fazil et al., [10] conducted a study in Afghanistan's Badakhshan Province examining the impact of cybersecurity education on students' online safety and digital literacy. This study underlines the importance of instilling awareness of responsible digital practices concerning privacy, security, and copyright and calls for active parental involvement. The study's results revealed that the need for cybersecurity education among youth is becoming more pronounced every day [10]. In New Zealand, Tirumala et al.,

[30] conducted a survey on internet usage and cybersecurity awareness among students aged 8–21. The results showed that most students used the internet for entertainment. In terms of access, students were relatively more flexible in granting access to various applications, including personal details and privacy information, which makes this group more vulnerable to dangers (Tirumala et al., [30]).

Corradini and Nardelli (2020) [6] present a study investigating Italian school teachers' perceptions of their students' digital awareness and their assessment of the actions needed to improve it. The responses were provided by 2,229 teachers from primary and secondary schools across the country who participated in a national project aimed at popularizing computer science and raising students' sensitivity to the proper use of digital technologies. The results confirm the teachers' high sensitivity to digital awareness issues. Indeed, students should be prepared to recognize risks when using digital technologies. They should pay more attention not only to cyberbullying but also to the protection of their personal data and the reliability of news on social media.

International practices set an example for integrating cybersecurity education. Countries such as the United States, Singapore, and Japan have taken significant steps in this direction. For example, the United States, through the National Initiative for Cybersecurity Education (NICE) led by the National Institute of Standards and Technology (NIST), focuses on building a robust cybersecurity workforce. The NICE Framework provides a common lexicon to describe cybersecurity work, which is used by employers, educators, and training providers to develop curricula and career pathways [20].

Similarly, Singapore's Cyber Security Agency (CSA) has implemented the SG Cyber Safe Students Programme to raise awareness and promote good cyber hygiene among young people. While their official website details the program's objectives and initiatives, scholarly articles and government reports often analyze its effectiveness in practice [8].

Japan has incorporated cybersecurity into its educational systems, highlighting the global need for such programs [15]. These global efforts have become even more urgent due to recent events. Al-Fatlawi, [2] revealed that the rapid shift to online learning during the COVID-19 pandemic created new security vulnerabilities for both students and educators. This reinforces the idea that cybersecurity education should not be a one-time event but a continuous and comprehensive part of the curriculum [36].

Cybersecurity education should be seen as an indispensable component of modern education, equipping young students with the tools and knowledge to interact with technology safely and responsibly. It is crucial to raise a new generation of knowledgeable digital citizens who can navigate the digital world responsibly and securely [7]. As emphasized by Rahman et al., [24], cybersecurity education is mandatory to address the evolving threats in the digital age. It is now more important than ever for students to be equipped with the knowledge and skills to protect themselves and their information online. Cybersecurity education not only prevents individuals from becoming victims of cyber threats, but it also cultivates a sense of responsibility and ethical behavior.

3 Methods and findings

This study used quantitative research techniques to measure cybersecurity awareness. A five-point Likert-type survey on cybersecurity concepts was distributed as an online survey to pilot high schools selected from the same district. Although the high schools were located in the same district, they were selected from different types to ensure

demographic diversity and representativeness. Due to ethical guidelines and institutional confidentiality, the names of the schools have been withheld. The school types are as follows: the Anatolian High School, the Girls' Anatolian Imam Hatip High School, the Vocational and Technical Anatolian High School, the Science High School. At the end of the survey, participants were asked to complete a short quiz. Participants were asked to judge whether each of 10 sample website screenshots was genuine or fraudulent.

The fraudulent-screen task was designed as follows. The scenario visuals were developed from real-world attack cases documented in the literature and from publicly reported incidents. Each scenario deliberately included cues that could help participants judge whether a page was fraudulent, such as unusual or urgent requests, mismatches between displayed URLs and embedded hyperlinks, grammatical and spelling errors, and abnormal visual design or formatting. Of the ten screens, eight were fraudulent and two were genuine (Screens 1 and 6). A higher proportion of fraudulent screens was used so that participants would be exposed to a range of attack patterns, while the genuine screens were included to discourage participants from assuming that every page was an attack and thus to reduce response bias. The screens were presented in a fixed order within a single online survey and participants were not told how many of the screens were fraudulent. For each screen, participants classified the page as genuine or fraudulent. A response was scored as correct when a fraudulent page was identified as fraudulent and a genuine page as genuine.

All 10 screens were analyzed with McNemar tests and reported in Table 5. Four of the screens are shown as examples in the Appendix B, including three fraudulent screens (Figs. 2, 4 and 5) and one genuine screen (Fig. 3).

This study aimed to check for any change in students' awareness levels after receiving cybersecurity training by conducting the training following the initial survey. To obtain two separate data sets, pre and post-training, the same survey was sent to all four high schools. The first survey responses were obtained between November and December 2024. Cybersecurity training was then provided to the four pilot high schools between February and March 2025. Students were called to the conference halls in their respective schools and attended a standardized, one-hour training session delivered by instructors from the Department of Management Information Systems at Gebze Technical University, including Research Assistants and Associate Professors with expertise in information systems and cybersecurity-related topics. All four high schools completed the same training using the same core instructional materials and learning objectives. The training was designed as an introductory session for students with no assumed prior cybersecurity knowledge and included basic introductions to cybersecurity, theoretical cybersecurity terminology, personal security measures, and practical guidance on how to identify fraudulent websites and misleading online content. Emphasis was placed on the idea that not every screen, website, or piece of information encountered on the Internet is authentic or trustworthy. To support these concepts, the training incorporated examples, visual demonstrations, and selected YouTube videos related to cybersecurity. The one-hour session consisted of an introduction to cybersecurity, presentation of the core concepts and terminology, discussion of personal security measures, demonstrations and video-based examples of fraudulent or misleading websites and online content, and interaction with students through questions and discussion. Although the same core content and materials were used across all four schools, minor differences in

discussion could arise based on students' questions and the school context. If requested, training materials and the links of the videos can be shared with other researchers or educators. After the training, the same survey was sent to the same high schools again and responses were collected in April 2025.

A total of 1081 responses were received before and 881 after the training, respectively. Since the study aimed to focus solely on students, teachers' responses were removed from the dataset, leaving 1029 student data points before the training and 870 after the training. For the pre-post comparison, only students who completed both surveys and could be matched by school and student number were kept, which produced a final matched sample of 601. Analyses were performed on this dataset of paired responses from the same students before and after the training so that individual-level change could be examined directly.

Table 1 presents the demographic distribution by school, grade, and gender.

To assess potential attrition bias, matched ($N = 601$) and unmatched ($N = 428$) students from the pre-training wave were compared on gender, school, and grade using chi-square tests. The groups differed on all three characteristics (gender: $\chi^2(1) = 4.63$, $p = .031$; school: $\chi^2(3) = 268.83$, $p < .001$; grade: $\chi^2(5) = 104.71$, $p < .001$). Matching rates ranged from 15.6% to 86.7% across the four schools.

The validity of the following hypotheses was tested within the scope of the research.

H1: Before the training there is a significant difference in cybersecurity awareness levels between different schools.

H2: Before the training, participants' cybersecurity awareness differs significantly based on their grades.

H3: Before the training, there is a significant difference in cybersecurity awareness levels based on the gender of the participants.

H4: After the training, there is a significant difference in cybersecurity awareness levels between different schools.

H5: After the training, participants' cybersecurity awareness differs significantly based on their grade level.

Table 1 Demographic characteristics of the matched sample used in the analysis ($N = 601$)

Variables	Category	Frequency (*n*)	Percentage (%)
Gender	Female	373	62.1
	Male	228	37.9
School Type	Anatolian High School	281	46.8
	Science High School	157	26.1
	Girls' Anatolian Imam Hatip High School	130	21.6
	Vocational and Technical High School	33	5.5
Grade Level	9th Grade	228	38.1
	10th Grade	182	30.4
	11th Grade	149	24.9
	12th Grade	40	6.7
Missing	Grades	2	

H6: After the training, there is a significant difference in cybersecurity awareness levels based on the gender of the participants.

H7: There is a statistically significant change in participants' awareness levels following the cybersecurity training.

All statistical analyses were performed using IBM SPSS Statistics software (Version 26.0).

To test the construct validity of the scale, pre-training responses were used. Teacher responses were set aside so that the analysis reflected the student population the study focuses on, which left 1029 student responses. The factor structure was established on this pre-training sample, while the pre- and post-training comparisons reported later were conducted on the matched sample of students who completed both measurements ($N = 601$). Principal Component Analysis (PCA) was used to examine the underlying structure of the scale and the suitability of the data for this analysis was first assessed. The Kaiser-Meyer-Olkin (KMO) sample adequacy criterion was obtained with a score of 0.905, indicating the suitability of the data. The Bartlett Sphericity test was applied to measure the suitability for factor analysis. Statistically significant results were obtained at the end of the test ($\chi^2 (210) = 6508.127, p < .001$). Varimax rotation with Kaiser normalization was applied to improve the interpretability of the factor structure.

The number of factors was determined by considering the Kaiser-Guttman criterion (eigenvalues > 1), [12], the scree plot and the conceptual interpretability of the factors together. While the scree plot, presented in Fig. 1, suggested a smaller number of components, the five-factor solution was retained because it was conceptually interpretable and consistent with the theoretical framework of the study. Collectively, these five factors, with initial eigenvalues of 6.24, 2.07, 1.30, 1.15, and 1.08 respectively, accounted for 56.42% of the total variance. The rotated component matrix and the communalities for all items are presented in Table 2. The conceptual themes and underlying structures of the five extracted factors are detailed below.

One of the 22 items in the survey was excluded from the analysis because it was an open-ended, non-Likert-type item asking about daily internet usage time (Q1). The

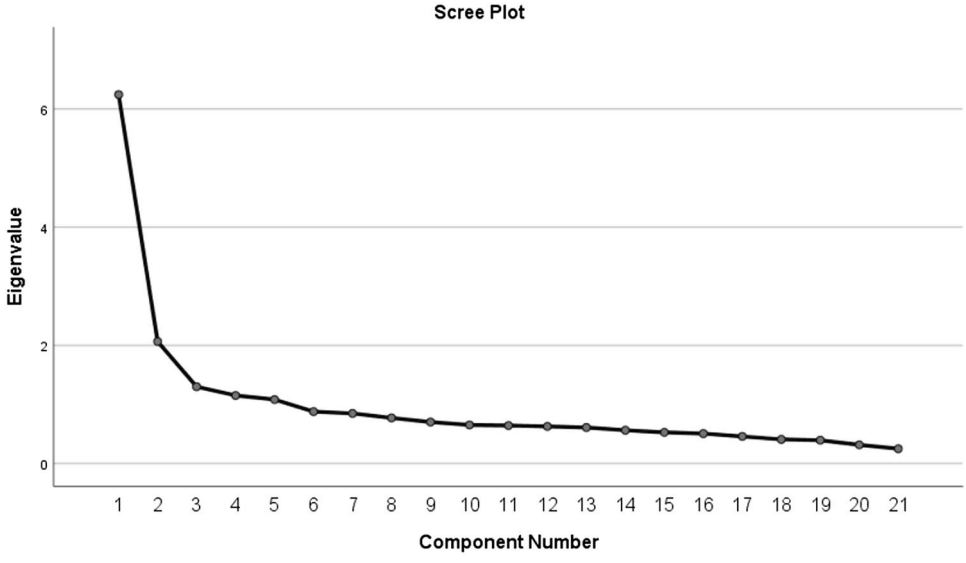


Fig. 1 Scree plot of the principal component analysis

Table 2 Rotated component matrix and communalities ($N = 1029$)

Item	F1	F2	F3	F4	F5	h^2
Q9	**0.824**	0.146	0.161	−0.021	−0.008	0.726
Q10	**0.771**	0.121	0.260	0.011	−0.028	0.677
Q7	**0.763**	0.126	0.199	0.003	0.084	0.645
Q6	**0.655**	0.189	−0.082	0.173	0.142	0.522
Q3	**0.579**	0.205	0.312	−0.110	0.018	0.487
Q8	**0.567**	0.108	**0.529**	0.068	−0.027	0.619
Q11	**0.535**	0.158	0.403	−0.001	−0.159	0.499
Q13	0.420	0.418	−0.076	0.305	−0.156	0.474
Q17	0.103	**0.828**	0.100	−0.082	−0.125	0.728
Q18	0.134	**0.819**	0.025	−0.037	−0.092	0.699
Q16	0.197	**0.552**	0.055	0.265	0.209	0.460
Q22	0.276	**0.505**	0.050	0.370	0.057	0.474
Q15	0.101	**0.484**	0.142	0.115	0.212	0.323
Q14	0.286	**0.405**	0.241	0.301	−0.210	0.439
Q5	0.150	0.016	**0.789**	0.235	−0.019	0.701
Q4	0.220	0.088	**0.767**	−0.085	0.073	0.656
Q2	0.366	0.305	**0.446**	−0.335	−0.009	0.538
Q21	−0.052	0.184	0.095	**0.762**	0.044	0.629
Q19	0.208	0.348	0.341	**0.354**	0.005	0.406
Q12	0.010	−0.021	−0.049	0.057	**0.817**	0.674
Q20	−0.055	−0.077	−0.120	**0.462**	**−0.483**	0.471

Extraction method: Principal Component Analysis. Rotation method: Varimax with Kaiser normalization. $N = 1029$. Loadings ≥ 0.40 are shown in bold. h^2 = communality (extraction)

remaining 21 items were included in the Principal Component Analysis, and all 21 items were retained under the five factors without any items being removed due to low loading or cross-loading. Two behavioural items (Q12, Q20) were reverse-coded before analysis so that, across all items, a higher score consistently indicates safer behaviour. The full item wording and response options are provided in Appendix A.

Each item was assigned to the factor on which it showed its highest loading. A small number of items (for example, the item on precautions when using public Wi-Fi and the item on antivirus software) showed comparable loadings on a second factor. These cross-loadings are reported transparently in the factor table and the items were retained on their highest-loading factor, since this assignment was also consistent with the conceptual content of each item. Each factor score was then computed as the mean of the items assigned to that factor.

Factor 1 (Conceptual Awareness of Cybersecurity): This dimension contains the questions Q3, Q6, Q7, Q8, Q9, Q10, Q11, and Q13 and reflects students' knowledge of fundamental cybersecurity concepts and terminology. Focusing on theoretical knowledge such as malware, firewalls, and social engineering rather than actionable behaviors, this dimension assesses the cognitive foundation required to recognize digital threats, understand connection security, and comprehend the functions of basic defensive software.

Factor 2 (Action-Oriented Behaviors): Factor 2 contains Q14, Q15, Q16, Q17, Q18, Q22 and focuses on action-oriented security habits applied in daily digital routines. This dimension evaluates the extent to which passive cybersecurity knowledge is transformed into active personal measures. The steps students take regarding device authentication, email checking, monitoring current risks, and account and password security were examined.

Factor 3 (Awareness of Advanced Protection Mechanisms): Factor 3, which includes questions Q2, Q4, and Q5, examines the level of understanding of structurally more advanced privacy and authentication tools. By measuring awareness of secondary defense layers, such as VPN, two-factor authentication logic and the necessity of strong passwords, this dimension represents the transition between recognizing cyber threats and knowing the specialized tools designed to protect against them.

Factor 4 (Security in Online Interactions and Transactions): Factor 4, encompassing Q19 and Q21, addresses the socio-technical aspect of cybersecurity by focusing on how participants manage their personal boundaries and assets during their active online engagement. The grouping of the cyberbullying and card-security items suggests that this factor relates to security behaviors in digital public spaces, combining transactional awareness with interpersonal digital security. This subscale consists of only two items with low internal consistency. Its interpretation is therefore treated with caution.

Factor 5 (Digital Risk Tendency and Privacy Management): Factor 5 includes Q12 and Q20. It captures risk taking tendencies in everyday or unsupervised digital environments. Considering habits of connecting to unsecured networks and the propensity to share credentials in online gaming or live streaming, this dimension measures how users manage their data privacy in entertainment focused online activities and their perception of risk in unprotected networks. After reverse-coding, a higher score on this dimension reflects a lower risk tendency and more privacy-conscious behaviour.

Along with the examination of the factor structure, the internal consistency of the scale was examined. The overall scale showed acceptable internal consistency (Cronbach's alpha = 0.86). Because the instrument is multidimensional, reliability was also examined separately for each subscale (Table 3). For the multi-item factors Cronbach's alpha was used and for the two-item factors the Spearman-Brown coefficient was reported as it is more appropriate for two-item scales. Although Factor 3 emerged with three items in the factor solution (Q2, Q4, and Q5), its composite score was computed from Q4 and Q5 only. The two-factor-authentication item (Q2) was retained in the factor solution but excluded from the score because its inclusion lowered the internal consistency of the subscale (three-item alpha = 0.493 versus two-item Spearman-Brown = 0.597). The multi-item factors showed acceptable values, whereas the two-item Factor 4 and Factor 5 showed lower coefficients and are therefore interpreted with caution.

Dependent samples t-tests were performed on the paired dataset (N = 601). Matching the pre- and post-training responses of the same students allowed within-participant change to be examined directly. However, since the study did not include a control group it does not remove attrition bias, testing effects or other time-varying influences. If a significant difference was found, Cohen's d value was calculated to assess the effect size of the training [5]. The results obtained from the analyses performed on five factors are shown in Table 4.

Table 3 Internal consistency reliability of the five subscales

Subscale	Number of Items	Reliability Coefficient
Factor 1: Conceptual Awareness of Cybersecurity	8	Cronbach's α = 0.853
Factor 2: Action-Oriented Behaviors	6	Cronbach's α = 0.762
Factor 3: Awareness of Advanced Protection Mechanisms	2	Spearman-Brown = 0.597
Factor 4: Security in Online Interactions and Transactions	2	Spearman-Brown = 0.365
Factor 5: Digital Risk Tendency and Privacy Management	2	Spearman-Brown = 0.110

A significant decrease was observed for Factor (1) A similar but more pronounced significant decrease was detected for Factor 3. For Factor 5, a small but statistically significant decrease was observed although the effect was minimal and this subscale should be interpreted with caution given its low internal consistency. In contrast, a statistically significant increase was measured in Factor (2) Finally, no statistically significant difference was found for Factor 4.

The decreases in Factors 1 and 3 can be read in more than one way. One reading draws on the Dunning-Kruger effect [16]. Since both factors measure students' knowledge of cybersecurity terminology, students who had overestimated their familiarity with these concepts may have judged their own level more realistically after the training and reported lower scores. The study did not, however, measure actual conceptual knowledge apart from self-assessment, so the same decline could equally reflect response-shift bias, the instability of a short self-report measure or a real drop in reported confidence. We therefore treat the more-realistic-assessment reading as one plausible interpretation rather than a proven one and do not present it as direct evidence of improved threat appraisal.

The small increase in Factor 2 suggests that students reported applying personal security steps somewhat more after the training. It appears that the training encouraged students to adopt protective measures. Unlike Factors 1 and 3, which contain technical information, Factor 2, which covers applicable practical measures, can be said to have been more easily adopted by students. This increase corresponds to a strengthening of coping appraisal in Protection Motivation Theory, that is, students' growing belief that protective measures are useful and within their reach.

For Factor 4 no significant change was observed (d = -0.04), indicating that scores remained essentially stable from before to after the training. This dimension reflects more established social attitudes which may be less responsive to a short-term intervention.

3.1 Significant difference tests

This section of the study examined whether students' cybersecurity awareness levels differed according to demographic variables before and after the training. For this purpose, school, grade and gender variables were considered as independent variables, while the pre- and post-training scores of the five basic factors constituting the scale were used as dependent variables. Independent Samples t-test was performed for comparisons

Table 4 Paired-samples t-test results and effect sizes for cybersecurity awareness dimensions

Sub-dimensions	Pre (M ± SD)	Post (M ± SD)	t (df = 600)	*p*	Cohen's d [95% CI]	Mean diff. 95% CI
Factor 1: Conceptual Awareness of Cybersecurity	3.10 ± 1.05	2.44 ± 0.96	15.30	< 0.001	0.62 [0.54, 0.71]	[0.57, 0.74]
Factor 2: Action-Oriented Behaviors	3.05 ± 0.88	3.16 ± 0.92	-3.60	< 0.001	-0.15 [-0.23, -0.07]	[-0.17, -0.05]
Factor 3: Awareness of Advanced Protection Mechanisms	4.42 ± 0.78	1.85 ± 0.98	45.42	< 0.001	1.85 [1.72, 1.98]	[2.47, 2.69]
Factor 4: Security in Online Interactions and Transactions	3.54 ± 0.91	3.58 ± 0.93	-0.95	0.344	-0.04 [-0.12, 0.04]	[-0.12, 0.04]
Factor 5: Digital Risk Tendency and Privacy Management	4.27 ± 0.76	4.14 ± 0.96	3.36	0.001	0.14 [0.06, 0.22]	[0.06, 0.21]

Values are based on the pre-minus-post difference. A positive Cohen's d and t indicate a decrease from pre- to post-training, and a negative value indicates an increase

between two groups and One-Way Analysis of Variance (ANOVA) was performed for comparisons between more than two groups. As a prerequisite before ANOVA analyses, the homogeneity of variances was checked with the Levene Test. If the variances were homogeneous the Tukey HSD test was performed to find the source of the difference in the analyses. In cases where homogeneity was not found, the Welch test was considered for the equality of means and the Games-Howell post-hoc analysis was preferred.

3.1.1 Pre- and post-training analysis by school types

According to this analysis framework, the analysis was first performed according to schools.

A one-way ANOVA indicated that Factor 1 pre-training awareness scores differed significantly across school types, $F(3, 597) = 8.84$, $p < .001$, $\eta^2 = 0.04$. Because the homogeneity of variance assumption was met (Levene $p = .226$), the Tukey HSD test was used to identify which schools differed. According to the Tukey HSD test result, the pre-education cyber threat awareness scores of the Girls' Anatolian Imam Hatip High School students were statistically significantly lower than those of the Anatolian High School ($p = .030$), Vocational and Technical Anatolian High School ($p = .002$), and Science High School ($p < .001$). This indicates that these three schools formed a group with similar high awareness before education, while the Girls' Anatolian Imam Hatip High School scored significantly lower than this group.

When the changes in Factor 1 scores across schools after the training were analyzed, it was concluded that the data met the assumption of homogeneity of variance (Levene $p = .117$), however, the ANOVA test results showed no significant difference between schools for Factor 1 ($F(3, 597) = 1.66$, $p = .175$, $\eta^2 = 0.01$). After the training, the school means were close to one another (Anatolian High School M = 2.47; Girls' Anatolian Imam Hatip High School M = 2.47; Vocational High School M = 2.29; Science High School M = 2.38). Because the study included no control group and no time-by-school interaction was tested, the significant pre-training difference and the non-significant post-training result are described as observed differences at each time point rather than as evidence that the training equalized the schools.

When the analysis was performed with pre-education scores for Factor 2, homogeneity of variance was determined (Levene $p = .430$). According to the ANOVA result, a significant difference was found between the schools ($F(3, 597) = 6.47$, $p < .001$, $\eta^2 = 0.03$), so the group causing the difference was investigated. According to the results of the Tukey HSD post-hoc test performed for this purpose, the initial scores of Anatolian High School students (M = 3.17) were statistically significantly higher than those of Science High School students (M = 2.86) ($p < .001$). While descriptive statistics showed that Vocational and Technical Anatolian High School (M = 3.20) had the highest arithmetic mean, multiple comparison tests showed that there was no statistically significant difference between this school and the other groups. According to the results obtained, it was observed that there was a significant difference in theoretical background only between Anatolian High School and Science High School in the Factor 2 dimension before education, while other institutional structures shared a similar initial level.

In the Factor 2 dimension, the homogeneity of variances was first checked with the post-training scores. According to the Levene test result (Levene $p = .002$), it was seen that the assumption of homogeneity of variances was not met. Therefore, the Welch

ANOVA test, which is more robust against variance inequalities, was used. According to the Welch test result, it was understood that a statistically significant difference continued between school types even after the training (Welch $F(3, 132.38) = 3.27$, $p = .023$, $\eta^2 = 0.02$). Due to the non-homogeneity of variances, the Games-Howell test, a post-hoc analysis, was performed to determine the source of the significant difference between the groups. According to the findings, as before the training, the Factor 2 scores of Anatolian High School students ($M = 3.25$) were statistically significantly higher than those of Science High School students ($M = 3.01$) after the training ($p = .041$). As before the training, the gap between Anatolian High School and Science High School students in the Factor 2 dimension remained significant after the training.

For Factor 3 pre-training scores, Levene's test showed that the group variances were not equal ($p = .004$), so the Welch ANOVA was used. Scores differed across schools (Welch $F(3, 133.17) = 4.06$, $p = .008$, $\eta^2 = 0.02$). Games-Howell comparisons showed that students at the Girls' Anatolian Imam Hatip High School ($M = 4.23$) scored lower than students at the Vocational and Technical High School ($M = 4.65$, $p = .045$) and the Science High School ($M = 4.53$, $p = .012$), while the difference from the Anatolian High School ($M = 4.42$) was not significant ($p = .142$).

After the training, Factor 3 scores again differed across the four schools (Levene $p = .059$; $F(3, 597) = 3.20$, $p = .023$, $\eta^2 = 0.02$). Tukey comparisons showed that the only significant difference was between the Girls' Anatolian Imam Hatip High School ($M = 2.00$) and the Vocational and Technical High School ($M = 1.42$, $p = .014$). The ordering of the schools changed from before to after the training: the Girls' Anatolian Imam Hatip school, which had the lowest pre-training mean, had the highest post-training mean, while the Vocational and Technical school moved from the highest to the lowest. Because the study did not include a control group and no time-by-school interaction was tested, these are described as observed changes in the group means rather than evidence of a levelling effect.

Homogeneity of variance was detected in the pre-education scores for Factor 4 (Levene $p = .702$), and a statistically significant difference was found between school types ($F(3, 597) = 3.26$, $p = .021$, $\eta^2 = 0.02$). Further analysis revealed that this difference stemmed from the fact that the pre-education scores of the Girls' Anatolian Imam Hatip High School students ($M = 3.71$) were statistically significantly higher than those of the Science High School students ($M = 3.41$) ($p = .024$).

When examining whether the Factor 4 post-test scores of the participants differed according to school type after the training intervention, the Levene test results showed that the data did not satisfy the assumption of homogeneity of variance (Levene $p = .014$). According to the Welch test results, no statistically significant difference was found between school types after the training (Welch $F(3, 132.56) = 1.15$, $p = .331$, $\eta^2 = 0.01$). The pre-training difference, in particular the gap between the Science High School and the Girls' Imam Hatip High School, was significant, whereas the post-training comparison was not (Welch $p = .331$). Descriptive statistics indicate that the groups largely maintained their initial scores with only a slight convergence towards the mean. Because no control group or time-by-school interaction was included this is described as an observed pattern rather than as evidence that the training equalized or maintained awareness levels for Factor 4.

For Factor 5, the pre-training scores were examined by school. Because Levene's test based on the mean indicated unequal variances (p = .006), the Welch test was used. Scores differed across schools (Welch F(3, 135.61) = 4.37, p = .006, η^2 = 0.02). In the Games-Howell comparisons, the only significant difference was between the Girls' Anatolian Imam Hatip High School (M = 4.44), which had the highest mean, and the Anatolian High School (M = 4.19, p = .003); the differences among the other schools were not significant.

After the training, Factor 5 scores again differed across schools (Levene p < .001; Welch F(3, 133.13) = 5.24, p = .002, η^2 = 0.02). The Girls' Anatolian Imam Hatip High School (M = 4.38) scored higher than both the Anatolian High School (M = 4.06, p = .001) and the Science High School (M = 4.07, p = .021), while the remaining comparisons were not significant. The Girls' Imam Hatip High School thus had the highest mean both before and after the training. Since the study included no control group, no time-by-school interaction was tested and Factor 5 showed low internal consistency, these results are described as observed differences in group means rather than as evidence of an effect of the training on the gap between schools and they are interpreted with caution.

3.1.2 Pre- and post-training analysis by grades

A series of analyses were conducted to determine whether the pre-education scores of the students participating in the study in the Factor 1 dimension differed according to their grade level. Although homogeneity of variance was detected (Levene p = .854), there was no statistically significant difference between the grade levels of the students and their Factor 1 pre-education awareness scores (F(3, 595) = 1.03, p = .380, η^2 = 0.01). When descriptive statistics were examined it was seen that the initial awareness of cyber threats among 9th grade (M = 3.12), 10th grade (M = 3.03), 11th grade (M = 3.19) and 12th grade (M = 3.01) students was quite close and at a similar level.

When the same analyses were performed with post-training scores, similar homogeneity of variance was detected (Levene p = .082), but no significant differences were found between grades (F(3, 595) = 0.63, p = .594, η^2 < 0.01). According to descriptive statistics, it was observed that the mean scores of 9th grade (M = 2.47), 10th grade (M = 2.43), 11th grade (M = 2.41) and 12th grade (M = 2.36) students were similar after the training. Grade-level means were similar both before and after the training. Because no time-by-grade interaction was tested, this is described as an observed similarity across grades rather than as evidence that the training affected the grades equally.

Analyses conducted to measure whether Factor 2 scores differed according to grades revealed homogeneity of variance before the training (Levene p = .542), but there was no statistically significant difference between the pre-training scores of 9th, 10th, 11th, and 12th grade students (F(3, 595) = 1.18, p = .315, η^2 = 0.01). When the post-training analysis results were examined, it was determined that the assumption of homogeneity of variance was not met (Levene p = .019). Accordingly, according to the results of the Welch test, there was no statistically significant difference between grade levels after the training (Welch F(3, 170.25) = 2.26, p = .083, η^2 = 0.01). For Factor 2, grade-level scores did not differ significantly either before or after the training.

When the Factor 3 scores were examined by grade level, the Levene test based on the mean showed that the variances were not homogeneous across grades before the training (p = .034), so the Welch test was used. The awareness scores of 9th grade (M = 4.45),

10th grade (M = 4.46), 11th grade (M = 4.40), and 12th grade (M = 4.25) students were close to one another, and no statistically significant difference was found between grade levels (Welch F(3, 157.77) = 0.70, p = .555, $\eta^2 < 0.01$). None of the Games-Howell pairwise comparisons was significant, and all four grades fell within a single homogeneous subset. After the training, the assumption of homogeneity of variance was met (Levene p = .121), so the standard ANOVA was used. Again, no significant difference was found between grade levels (F(3, 595) = 1.68, p = .171, $\eta^2 < 0.01$), with the grade means remaining close (9th grade M = 1.91, 10th grade M = 1.85, 11th grade M = 1.71, 12th grade M = 2.04). In line with the other dimensions, the absence of a difference both before and after the training indicates that for Factor 3, awareness did not vary systematically by grade level.

Similarly, when Factor 4 scores were examined homogeneity of variance was observed before and after the training (Levene p = .676; Levene p = .459) but no significant difference was found between grades in either time period (F(3, 595) = 1.21, p = .304, $\eta^2 = 0.01$; F(3, 595) = 1.94, p = .122, $\eta^2 = 0.01$).

For Factor 5, the scores were examined by grade level. Before the training, Levene's test based on the mean indicated unequal variances (p = .003), so the Welch test was used. The result was marginal (Welch F(3, 161.12) = 2.76, p = .044, $\eta^2 = 0.01$) but no pairwise difference was significant in the Games-Howell test and all four grades fell within a single homogeneous subset, indicating that grade level was not meaningfully associated with Factor 5 scores. After the training no significant difference was found across grades (Levene p = .022; Welch F(3, 158.06) = 1.90, p = .132, $\eta^2 = 0.01$). Given the marginal nature of the pre-training result, the absence of any significant pairwise difference and the low internal consistency of this subscale, grade level is not treated as a meaningful source of variation in Factor 5.

The grade-level analyses were then considered as a whole. Across the five dimensions, grade level was not a meaningful source of variation. No pairwise differences between the 9th, 10th, 11th, and 12th grades were significant and the only omnibus result to reach significance (Factor 5 before the training) was marginal. It was not supported by any significant pairwise comparison and concerned a subscale with low internal consistency. Grade-level means were also similar before and after the training. Because no time-by-grade interaction was tested, these patterns are described as observed similarities across grades rather than as evidence that the training affected the grades equally.

3.1.3 Pre- and post-training analysis by gender

An independent samples t-test was used to analyze whether the students' Factor 1 levels differed according to gender. When pre-training data were examined, it was observed that the mean scores of male students (M = 3.46) were statistically significantly higher than those of female students (M = 2.87) (t_{599} = 8.04, p < .001, d = 0.68, 95% CI [0.45, 0.73]). However, in the post-test measurements taken after the cybersecurity training, this significant difference between genders disappeared (t_{599} = -1.62, p = .107, d = -0.14, 95% CI [-0.18, 0.02]). After the training, the scores of female (M = 2.47) and male (M = 2.39) students were close to one another. Because no time-by-gender interaction was tested, the significant pre-training difference and the non-significant post-training result are described as observed differences at each time point rather than as evidence that the training had an equalizing effect across genders.

The analysis of the Factor 2 levels of the students did not meet the assumption of homogeneity of variance (Levene test $p < .05$). When the pre-training data were examined, it was found that male students (M = 3.14) had statistically significantly higher awareness scores than female students (M = 2.99) ($t_{423.03} = 2.29$, $p = .023$, d = 0.20, 95% CI [0.02, 0.29]). A similar pattern was observed in the post-training measurements. Male students (M = 3.25) exhibited significantly higher scores than female students (M = 3.10) ($t_{405.99} = 2.14$, $p = .033$, d = 0.19, 95% CI [0.01, 0.30]). For Factor 2, male students scored higher than female students both before and after the training, and the difference remained significant at both time points.

Analyses at the Factor 3 level showed that, before the training, male students (M = 4.59) had statistically significantly higher awareness scores than female students (M = 4.32). Because the Levene test indicated unequal variances ($p = .006$), the equal-variances-not-assumed result was reported, $t_{518.54} = 4.25$, $p < .001$, d = 0.35, 95% CI [0.14, 0.39]. When the post-training data were examined the difference between genders remained significant but reversed in direction. Female students (M = 1.96) scored higher than male students (M = 1.66), $t_{599} = -3.73$, $p < .001$, d = -0.31, 95% CI [-0.47, -0.14]. Both groups' scores on this dimension declined overall. This reversal reflects a larger decrease among male students rather than an actual gain among female students.

When pre-training data were examined to compare Factor 4 awareness levels, it was found that female students (M = 3.65) had a statistically significantly higher initial awareness than male students (M = 3.36) ($t_{599} = -3.95$, $p < .001$, d = -0.33, 95% CI [-0.45, -0.15]). In post-training analyses, it was also observed that female students (M = 3.64) continued to score significantly higher than male students (M = 3.48) ($t_{599} = -2.02$, $p = .044$, d = -0.17, 95% CI [-0.31, 0.00]). Female students scored higher than male students both before and after the training and the difference was smaller in the post-training comparison. Because no time-by-gender interaction was tested, this narrowing is described as an observed change rather than as an effect of the training on the gap between genders.

For Factor 5, female students scored slightly higher than male students both before and after the training. Before the training, the difference was significant with equal variances assumed (Levene $p = .059$), $t_{599} = -2.17$, $p = .030$, d = -0.18, 95% CI [-0.26, -0.01] (female M = 4.32, male M = 4.19). After the training, with unequal variances (Levene $p < .001$), the difference remained significant, $t_{393.16} = -2.41$, $p = .017$, d = -0.22, 95% CI [-0.37, -0.04] (female M = 4.22, male M = 4.01). Both effects were small, and given the low internal consistency of this subscale, the gender difference on Factor 5 is interpreted with caution.

For the last part of this research's analysis the results of the McNemar test on the responses to the 10 screenshots shown at the end of the survey are given in Table 5. The ten McNemar tests were not adjusted for multiple comparisons and are treated as exploratory so the item-level results should be read with caution.

The results show a mixed pattern rather than a uniform increase in a single skill and they differ for the eight fraudulent screens and the two genuine screens. For the fraudulent screens, correct identification increased significantly for Screens 4, 7, 8, and 9, suggesting that the training helped students recognize several common deceptive designs. This improvement in practical recognition is consistent with the training functioning as a cue to action that strengthened students' response efficacy. No significant change

Table 5 McNemar test results for website screen identification

Screen	Type	Pre-test Correct, *n* (%)	Post-test Correct, *n* (%)	*p*
Screen 1	Genuine	509 (84.7%)	490 (81.5%)	0.110
Screen 2	Fraudulent	524 (87.2%)	517 (86.0%)	0.569
Screen 3	Fraudulent	523 (87.0%)	494 (82.2%)	0.017*
Screen 4	Fraudulent	303 (50.4%)	384 (63.9%)	< 0.001*
Screen 5	Fraudulent	374 (62.2%)	361 (60.1%)	0.441
Screen 6	Genuine	565 (94.0%)	539 (89.7%)	0.002*
Screen 7	Fraudulent	292 (48.6%)	350 (58.2%)	< 0.001*
Screen 8	Fraudulent	263 (43.8%)	329 (54.7%)	< 0.001*
Screen 9	Fraudulent	257 (42.8%)	321 (53.4%)	< 0.001*
Screen 10	Fraudulent	472 (78.5%)	436 (72.5%)	0.010*

Statistical significance is denoted by * for $p < .05$

was found for the fraudulent Screens 2 and 5. For Screen 2, 87.2% of students already answered correctly before the training, leaving little room for measurable improvement.

For the two genuine screens the pattern was different as the correct identification did not improve. For Screen 1 it edged down slightly but the change was not significant (84.7% to 81.5%, $p = .110$), and for Screen 6 it decreased significantly (94.0% to 89.7%, $p = .002$). On these genuine pages, students were somewhat more likely to label a legitimate page as fraudulent after the training. Correct identification also decreased for two of the fraudulent screens, Screens 3 and 10, both of which already had a high correct-identification rate before the training. Overall, the training was associated with better recognition of several fraudulent designs but not with a uniform improvement, and it did not improve recognition of the genuine pages. Future training could therefore give more attention to helping students distinguish fraudulent from genuine pages rather than treating every page as suspicious.

4 Conclusion

This study investigated the impact of a targeted cybersecurity education program on the awareness of high school students. Students from four different types of high schools in the same region were asked to participate in a cybersecurity awareness survey at two different times: before and after the training. The impact of the training on awareness change was examined using five factors. The findings showed that students reported applying more active protective measures while their self-reported scores on the conceptual dimensions declined. One possible interpretation of this decline is a shift toward more realistic self-assessment rather than a loss of knowledge, though this was not tested directly. Students became better at recognizing several of the fraudulent screens but their accuracy on the genuine screens did not improve and in one case declined, which suggests that future training should give more attention to distinguishing fraudulent from genuine pages rather than to suspicion alone.

On several dimensions, initial differences between schools were smaller after the training, although on others (for example Factor 3 and Factor 5) differences between schools persisted. Because no control group was used and no time-by-school interaction was tested, these patterns are described as observed changes rather than as an equalizing effect of the training. Within these limits standardized training covering the entire high school level may still be a more practical approach than curricula tailored to individual schools.

Grade level was not a meaningful source of variation in awareness. With one marginal exception, no significant differences emerged between the 9th and 12th grades, and grade-level means were similar before and after the training. This pattern suggests that a single programme spanning all high school grades may be a practical approach. Although, without a test of a time-by-grade interaction, this is offered as an observation rather than as evidence that the training affected all grades equally.

Gender-based analyses showed that, before the training, male students scored higher on the more knowledge-oriented dimensions (Factors 1, 2, and 3), while female students scored higher on the dimensions related to online interaction and privacy (Factors 4 and 5). After the training the gap on Factor 1 became smaller and the difference on Factor 3 reversed in favour of female students, although this reversal reflected a larger decline in male students' self-reported scores rather than a real gain. The overall pattern is consistent with observed shifts in the gender differences across dimensions rather than a uniform leveling effect, since some gaps became smaller while others persisted or slightly widened. Since no time-by-gender interaction was tested, these are described as observed changes rather than as an effect of the training on the gaps between groups.

Interpreted through the two models that framed the study, the training appears to have worked mainly as a cue to action in the sense of the Health Belief Model, prompting students to apply protective measures, while strengthening the coping appraisal described by Protection Motivation Theory. One way to read the mixed pattern in the self-reported scores is that practical performance and action-oriented behaviours improved, whereas self-assessments on the conceptual dimensions may have shifted toward a more realistic level. Because this interpretation was not tested directly, it is offered tentatively.

Overall, the results indicate that introducing structured cybersecurity education was associated with more action-oriented protective behaviors and with improved recognition on several of the fraudulent-page items while self-reported scores on the more conceptual dimensions declined, which may reflect a shift toward more realistic self-assessment. This underscores the importance of integrating cybersecurity awareness programs into secondary education and highlights the potential value of equipping young users with the skills needed to navigate digital environments safely and confidently. For future and more diverse research pilot schools from different districts can be selected to examine the potential influence of school location as well. Since children are starting to use the internet at increasingly younger ages, conducting a similar study in middle schools may help to examine the benefits of introducing cybersecurity education earlier.

5 Limitations

When interpreting the findings of this study, several limitations should be taken into account. The study used a single-group design in which the same students were measured before and after the training. This design did not include a comparison with students who received no training. Also, the schools consisted of a small number of pilot schools that were purposively selected from a single district and were not randomly assigned, so the results may not generalize to other contexts. Because each school type was represented by a single school, school-type effects cannot be separated from differences between individual schools. The findings therefore describe the four participating schools and should not be generalized to school types.

Participation levels differed before and after the training, reducing the dataset to 601 matched students. Comparisons between matched and unmatched students indicated that attrition was not random. Matching was associated with gender, school, and grade and matching rates were especially uneven across schools. The matched sample is therefore not fully representative of all students who participated at baseline and the findings should be interpreted with this attrition bias in mind. The small number of responses from the Vocational and Technical High School further reduces the stability of the between-school comparisons.

A further limitation concerns the composition of the sample in the gender comparisons. One of the participating schools was an all-girls Anatolian Imam Hatip high school, so a large share of the female respondents came from this single institution with its particular educational climate. As a result, the differences observed between male and female students cannot be cleanly separated from differences between school types, and the gender findings are partly confounded with school composition. Future studies could address this by drawing more balanced samples, for example by including both girls-only and boys-only schools, so that the gender effect can be examined more in isolation.

Because the same instrument was administered twice within a few months, part of the measured change may reflect growing familiarity with the questions rather than the training. In addition, awareness was measured largely through self-report which is open to social desirability and to inaccurate self-assessment. This point is important for interpretation since self-reported knowledge and performance on the practical fraudulent-page task did not always move in the same direction.

The analyses in this study involved many statistical tests across the five factors, the two time points and the school, grade and gender comparisons which increases the risk of chance findings. All of these comparisons were specified in advance as hypotheses (H1 to H7) rather than chosen after looking at the data so they are planned rather than post hoc. Even so, given their number and the absence of a correction for multiple comparisons the nominal p-values are read in a uniform and cautious way. Individual significant results, especially the smaller between-group differences and the item-level comparisons of the fraudulent-screen task, are treated as tentative rather than as firm evidence and more weight is placed on the overall pattern across analyses than on any single test.

The study also measured only short-term outcomes after the training and did not examine the durability of the change. In addition, Factor 4 and Factor 5 showed low internal consistency so the findings concerning these two dimensions are exploratory and should be interpreted with particular caution.

Supplementary Information

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Supplementary Material 1

Author contributions

C.C.C. conceived and designed the study. C.C.C., S.E., and F.H. developed the methodology and supervised the research process. B.O.K. and A.K. conducted data collection and implementation of the training program. F.H. and B.O.K. performed the data analysis. B.S. coordinated the project and facilitated collaboration with participating institutions. C.C.C. and F.H. wrote the main manuscript text, and all authors reviewed and approved the final manuscript.

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Data availability
The datasets generated during and/or analysed during the current study are not publicly available due to ethical and privacy considerations, as the data involve student participants and may contain potentially identifiable information, but are available from the corresponding author on reasonable request.

Declarations

Ethical approval
All procedures involving human participants were conducted in accordance with the ethical standards of the Gebze Technical University Human Research Ethics Committee and with the 1964 Declaration of Helsinki and its later amendments or comparable ethical standards. The study protocol was approved by the Gebze Technical University Human Research Ethics Committee (Approval No. 18 – 02). Informed consent was obtained from all individual participants included in the study. For participants under the age of 16, informed consent was obtained from their parents or legal guardians prior to participation.

Consent to participate
Informed consent was obtained from all individual participants included in the study. For participants under the age of 16, informed consent was obtained from their parents or legal guardians prior to participation.

Consent for publication
N/A.

Use of generative AI and AI-assisted technologies
During the preparation of this manuscript, the authors used a generative AI assistant (Claude, Anthropic) to help with English-language editing and translation to assist in drafting and refining parts of the text. All data collection and statistical analyses were carried out by the authors, who reviewed and edited every AI-assisted passage, verified all results, and take full responsibility for the content, analyses and conclusions of this work. No AI tool is listed as an author, in line with the journal's authorship criteria.

Competing interests
The authors declare no competing interests.

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